A-A-54327
5 February 1991

COMMERCIAL ITEM DESCRIPTION

DRILL, TWIST, BONE (Jacobs Chuck)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-D-42031(DM) which is cancelled.

This commercial item description covers Jacobs chuck bone twist drill suitable for bone drilling procedures.

Salient Characteristics:

The twist drills shall be of the following sizes as indicated below:

Size 1 - 1.1 mm diameter, 45 mm long
Size 2 - 1.5 mm diameter, 70 mm long
Size 3 - 2.0 mm diameter, 85 mm long
Size 4 - 2.7 mm diameter, 85 mm long
Size 5 - 3.5 mm diameter, 95 mm long

Material. The drills shall be fabricated of class 4, type 440B martensitic corrosion-resisting steel as specified in ASTM Designation F899 for composition only.

Style, design and dimensions. The drills shall be of one-piece construction. The proximal end of the drills shall be round and non-notched for use in Jacobs chuck type hand and power drills. The drills shall conform to Defense Personnel Support Center (DPSC) Drawing 23378 for design and dimensions, for the applicable size number. The flutes of the drills shall be uniformly cut, shaped, and sharpened to cut at the tip as well as the sides of the operative end and provide automatic chip clearance. Size 1 (1.1 mm) drill shall be suitable for drilling holes for a 1.5 mm screw. Size 2 (1.5 mm) drill shall be suitable for drilling holes for a 1.5 mm cortex screw and thread holes for a 2.0 mm cortex screw. Size 3 (2.0 mm) drill shall be suitable for drilling gliding holes for a 2.0 mm cortex screw and thread holes for 2.7, 3.5 and 4.0 mm cortex screws. Size 4 (2.7 mm) drill shall be suitable for drilling gliding holes for a 2.7 mm cortex screw. Size 5 (3.5 mm) shall be suitable for drilling gliding holes for a 3.5 mm cortex screw.

AMSC N/A FSC 6515

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Hardness. The drills shall have a Rockwell hardness of C56 to C58 in accordance with ASTM Designation E18 or E92 and E140.

Runout. The total indicated runout of the drills shall not exceed 0.001 inch. One drill of each size shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is utilized, the indicator shall be mounted with the shaft perpendicular to the axis of the drill, and so positioned that the foot contacts the drill at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the drill shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Sterilization. The drills shall be capable of withstanding repeated sterilization via steam, ethylene oxide or long term, 10 hour soaking in cold sanitization solution. The drills shall show no evidence of corrosion when autoclaved as specified herein.

Finish. The finish of the drills shall be smooth, bright and uniform, except the operative ends shall have a fine honed finish with sharp cutting edges.

Marking. The use of a trademark is acceptable. The type of marking or placement of the marking shall not adversely affect or interface with fit, function and operation of the drill with the Jacob chuck type hand and power drills. The marking shall be distinguishable to the unaided eye.

Instructions. Adequate and accurate instructions shall be furnished on the use and care of the twist drills. Instructions shall also include the type of solutions, both suitable and unsuitable for cold sanitization of the drills.

Workmanship. The drills shall free from defects such as cracks, pits, jagged edges or burrs which detract from their appearance or impair their serviceability.

Unit. Each (EA). One twist drill of the size specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Certificates of quality. Certificates of quality for raw materials, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection, examination, and test characteristics. The unit of product for inspection, examination, test purposes shall be one drill of each size. Unless otherwise specified, sampling for examination shall conducted at inspection level II, except for dimensions which shall be conducted at inspection level S-1, AQL 2.5 (percent defective) and sampling for tests (when applicable) shall be conducted at inspection level S-1. Acceptance number for tests shall be zero (0) for all samples.

Dimensions. Examination of the dimensions shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards that are used in conjunction with the gages.

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Tests. Tests shall be conducted to determine compliance with specification requirements.

Chemical analysis. In the absence of certificates of quality for the raw material, a chemical analysis shall be conducted. The chemical analysis shall be performed by reagent solutions, combustion methods, nonemission methods, or by spectrochemical analysis. Any ASTM method shall be acceptable.

Autoclaving. The drills shall be clean in accordance with the instructions supplied by the manufacturer. The drills shall then be autoclaved at 270° ± 5° F and 27 ± 1 psi for a minimum of 20 minutes.

Contract certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirement.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification	Size
6515-01-172-3229	DRILL, TWIST, BONE, 1.1 mm diameter, 45 mm long	1
6515-01-180-3231	DRILL, TWIST, BONE, 1.5 mm diameter, 70 mm long	2
6515-01-140-5189	DRILL, TWIST, BONE, 2.0 mm diameter, 85 mm long	3
6515-01-172-3228	DRILL, TWIST, BONE, 2.7 mm diameter, 85 mm long	4
6515-01-140-5190	DRILL, TWIST, BONE, 3.5 mm diameter, 95 mm long	5

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

MILITARY INTEREST:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6515-4666
LOCATION Word/drills

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54327	DRILL, TWIST, BONE (JaCobs Chuck)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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